

DQN, PG $\rightarrow$ implement on soccer game $\Rightarrow$ Find α^*

* policy we find is not a mixed strategy, "best mixed strategy" at each iteration

br dynamics

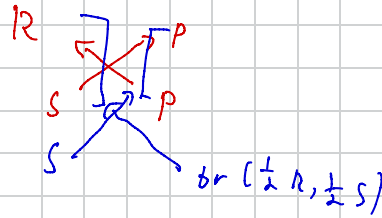
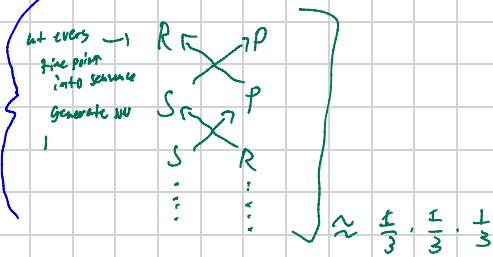
fictitious play

Does not converge to NE

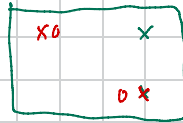
Side quest $\rightarrow$

always converges to Nash

Start



Symmetry - Swap x_1 & x_2 , are the activations the same

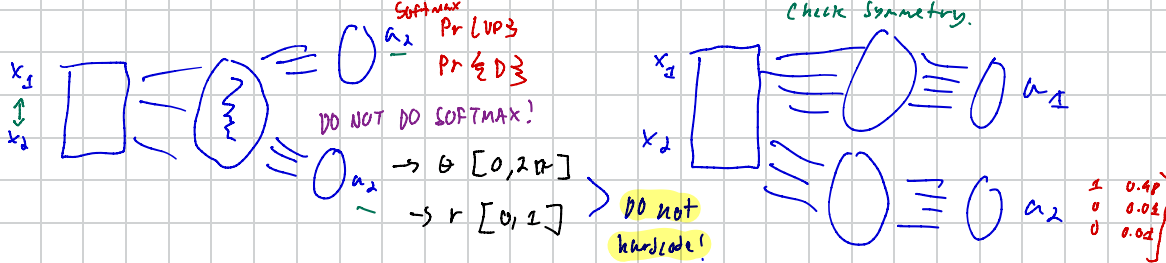


Respond to average of average decision.

Investigate this for next meeting

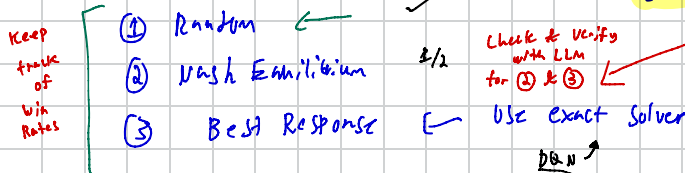
Reinforce All PPO

main quest: Comparing network architecture between players



Measure Performance

Get networks to converge!



Type of PG is your choice!!!

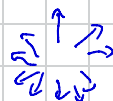
- If LLM makes assumption, tell it to let you know what the assumption is for each of those three players

big / dog

How to represent stage game & how to solve it



$Q(a_1)$
 $Q(a_2)$
 $Q(a_3)$



Problem: We need to avoid infinite # of actions

$\vdots$
 a_{10}

* Do not involve LP solver in these cases!

Go through basic optimization techniques we've seen before.

Put actions as networks, output Q

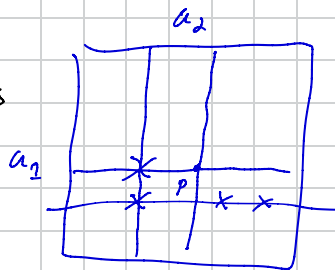


$Q(s_1, a_1, a_2)$

Needs to be a way to process Best Response Dynamics.

* Simple states & actions

$D = 2^{\text{tr}}$, so let network know. It has to be wrapped.



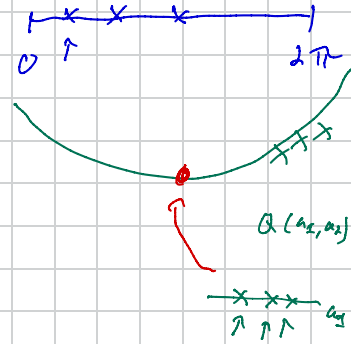
• Fix one player

• Use GD or some type of algorithm

Optimization Techniques

① bisection on derivative of Q (Q')

② gradient, $\frac{\partial Q}{\partial a}$
finite difference



Do finite difference to approximate GD!

③ Quadratic approximate

Sample function at any three points.

For different values of a_1 , find $Q(a_1, \bar{a}_2)$

LQR

(Linear Quadratic polynomials specifies games, and we can solve where Nash equilibria are more easily)

Side quest: Train a neural network to approximate a function where you know the global minimum



of times
 f is activated.

Once completed, train the b's/0's games.